University of Salford MANCHESTER

Advanced Web Development

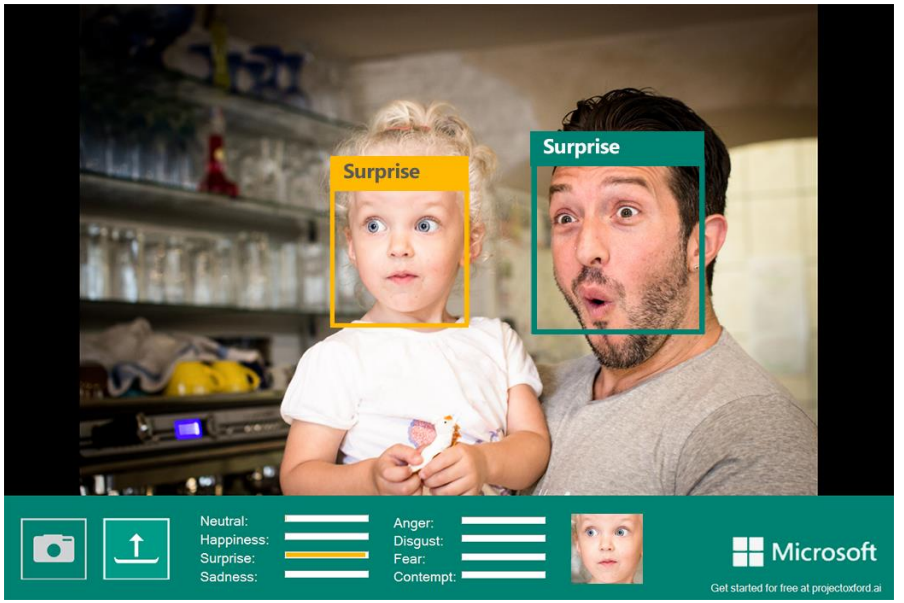
Introduction

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Web Development Trends

- Artificial Intelligence (face recognition, speech recognition, virtual assistants)
- Internet of Things
- Virtual / Augmented Reality







https://hackernoon.com/web-development-trends-2017-387421cf9c23

Outline

- Scope
- Eras in the History of the Web
- Key Concepts and Techniques
  - Basics
  - Web Programming
  - Design and Usability
  - Interoperability
  - Technological Evolution
  - Convergence of Desktop, Web, and Mobile Development
- Modern Web Development

Scope



- World Wide Web
  - Information space based on Hypertext/Hypermedia
  - Internet (an Intranet) as communication channel

- Web Development – tasks involved in web site creation
  - Content
  - Design and Artwork
  - **Technical Implementation / Software Development**

Eras in the History of the Web

- 1991 first web page – birth of the World Wide Web, invented by Tim Berners-Lee at CERN



- Static web pages
  - (predominantly linked texts and images)

- Dynamic web pages
  - (news portals, forums, wikis)





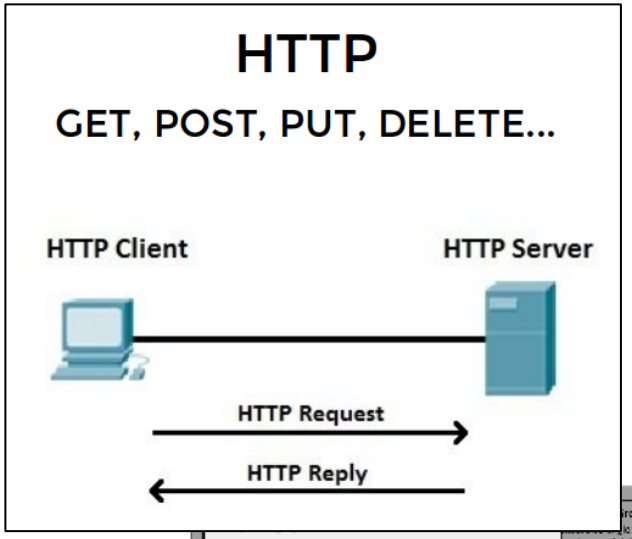
- Interactive web pages
  - (social media sites like Facebook)

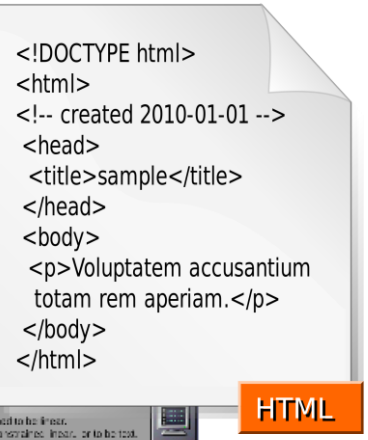


- Rich applications
  - (desktop-like software such as cloud-hosted Office editions)

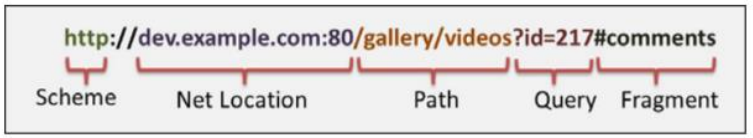
Basics

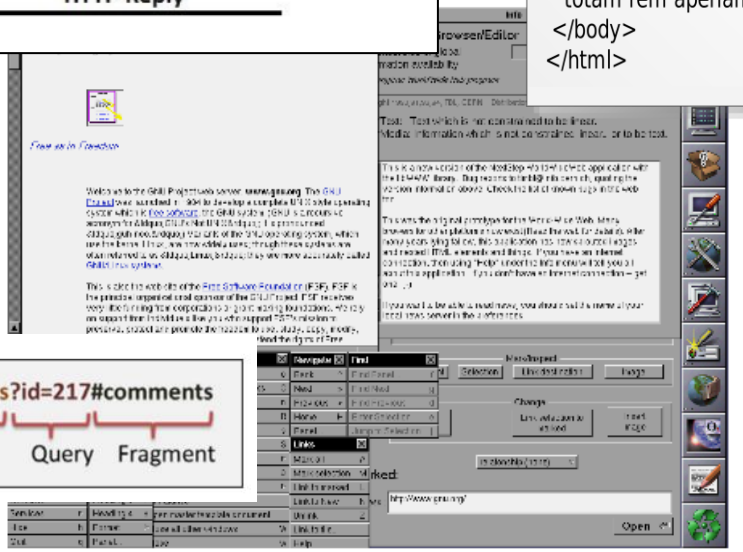
- Basic web ingredients:
  - HTTP
  - HTML
  - URL
  - Web Browser





- Client-server architecture
- Cross-platform approach



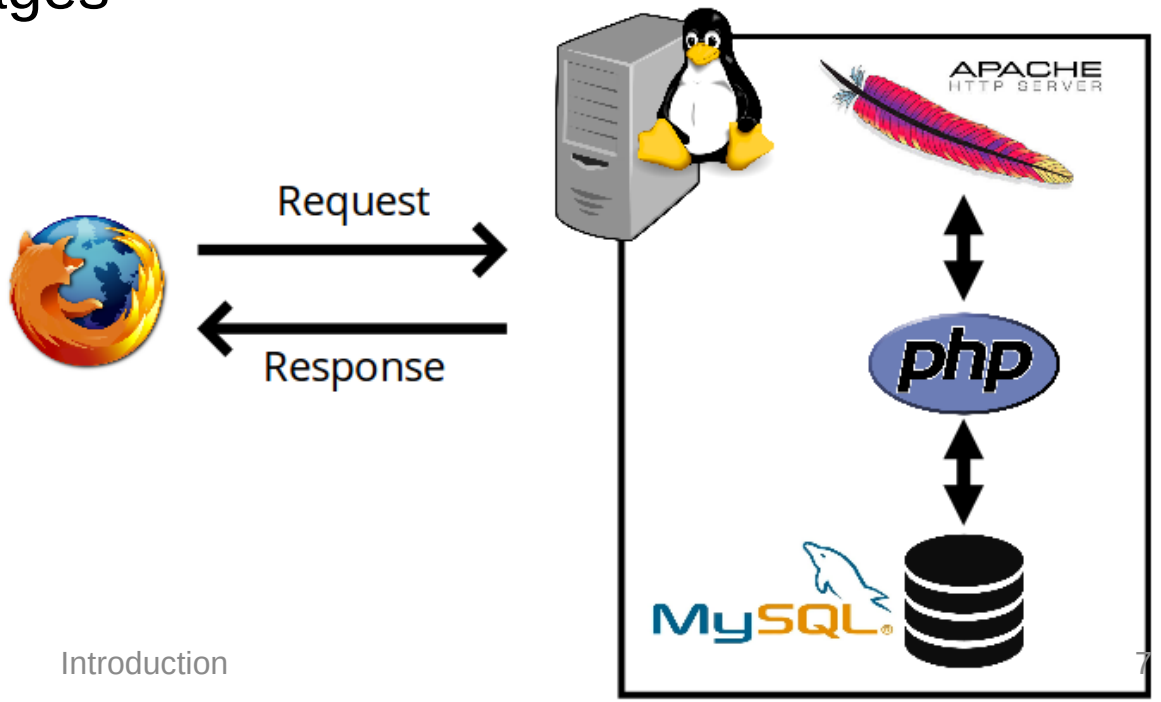


Web Programming

- Server-side programming
  - Dynamic generation of web pages
  - Database access
  - LAMP configurations, plethora of programming languages

- Client-side programming
  - Interactivity in web pages
  - DOM
  - JavaScript

- HTTP is stateless
  - Cookies
  - Sessions



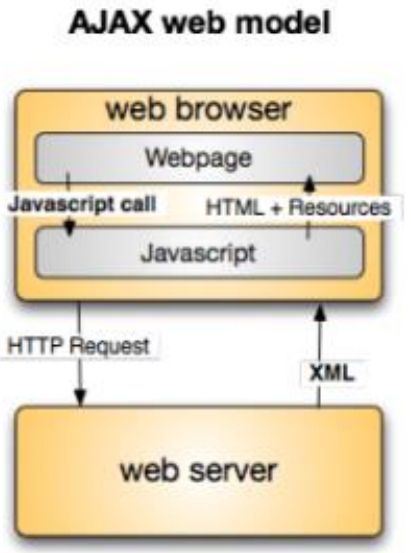
Design and Usability

- Separation of content and presentation
  - HTML + CSS

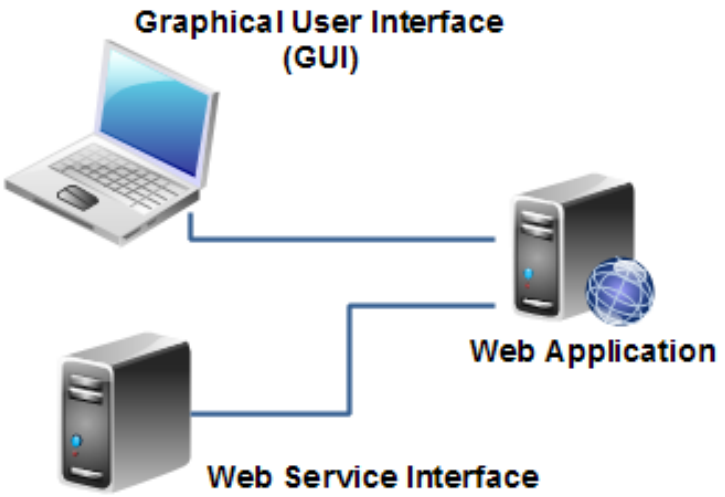


- Responsive Design
  - Accommodate different screen sizes, resolutions, and orientation
  - Bootstrap

- Asynchronous web applications
  - Updating of content without doing a full page reload
  - AJAX



Interoperability



- Formats for data exchange
  - XML
  - CSV
  - JSON
- Web services
  - Information exchange between systems (no front-end)
  - SOAP – protocol, services invoked via RPC
  - REST – architectural pattern, stateless, services called via URL
- WebSocket
  - Enables full-duplex connections (allows the server to call back)

Technological Evolution

- Next generation HTML
  - HTML5 + CSS3 + JavaScript
  - Improved multimedia capabilities (audio, video, and games)



- Server-side JavaScript
  - Node.js, V8



- Libraries and frameworks
  - Common client-side code
  - Back-end frameworks
  - Front-end frameworks
  - Full-stack frameworks













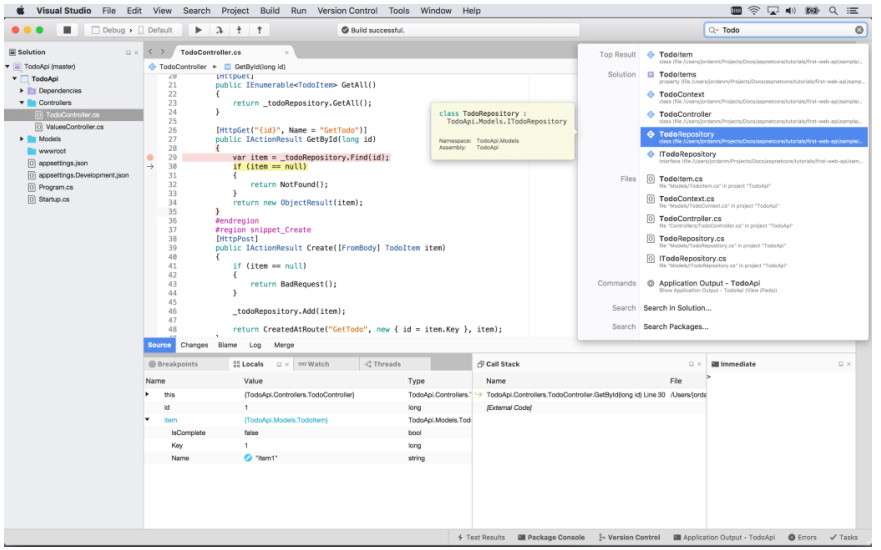






Convergence of Desktop, Web, and Mobile Development

- Desktop applications and Apps using web technologies
  - HTML5, CSS3, JavaScript
  - SPA - Single Page Application



- Unified development environments
  - Shared code base
  - MVC
  - Specifics of target platform hidden (to some degree)
  - E.g. Visual Studio, C#, and .NET

Modern Web Development

